IIT ROORKEE & FUTURENSE JOINT CERTIFICATION

Advanced PG Certification in AI Engineering on Cloud and AIOps

CAPSTONE FINAL PROJECT REPORT • THEME 15 (FINANCE DOMAIN)

AI Investment Research Assistant

Autonomous Multi-Agent Equity Research & Human-in-the-Loop Governance

SELECTED DOMAIN & THEME
Finance • Theme 15: AI Investment Research Assistant

ORCHESTRATION & FRAMEWORK
LangGraph DAG • ChromaDB RAG • Gemini Flash

COHORT DURATION
08 September 2026 – 16 October 2026

SUBMISSION DATE
16 October 2026

PROJECT TEAM MEMBERS & LEADERSHIP SPECIALIZATIONS

#	Member Name	Primary Engineering Domain & Specialization
1	**Ganesh Swami**	Backend Architecture, LangGraph DAG & Cloud DevOps Lead
2	**Shradha Gaikwad**	Financial Agent Intelligence & Quantitative Modeling Lead
3	**Sagar Trimukhe**	Frontend Dashboard & Chatbot/HITL Integration Lead
4	**Nilanjan Das**	Data Engineering, SEC Filings & Vector RAG Pipeline Lead
5	**Nikhil Gaikwad**	Full-Stack Integration, Telemetry & Quality Assurance Lead

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1. Project Title

AI Investment Research Assistant: Autonomous Multi-Agent Equity Research & Human-in-the-Loop Governance

- **Program:** Advanced PG Certification in AI Engineering on Cloud and AIOps
- **Institution:** Indian Institute of Technology (IIT) Roorkee, powered by Futurense
- **Cohort Duration:** 08 September 2026 – 16 October 2026
- **Submission Date:** 16 October 2026
- **Team Members:**
  1. **Ganesh Swami** — Backend Architecture & Cloud DevOps Lead
  2. **Shradha Gaikwad** — Financial Agent Intelligence & Quantitative Modeling Lead
  3. **Sagar Trimukhe** — Frontend Dashboard & Chatbot/HITL Integration Lead
  4. **Nilanjan Das** — Data Engineering, SEC Filings & Vector RAG Pipeline Lead
  5. **Nikhil Gaikwad** — Full-Stack Integration, Telemetry & Quality Assurance Lead

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2. Selected Domain and Theme

- **Domain:** Finance
- **Selected Theme:** Theme 15 — AI Investment Research Assistant
- **Target User Personas:**
  - **Buy-Side & Sell-Side Equity Analysts:** Requiring rapid financial statement extraction and peer multiple benchmarking across large corporate universes.
  - **Portfolio Managers:** Requiring quantitative volatility metrics, drawdown downside risk calculations, and synthesized investment theses before capital deployment.
  - **Investment Governance & Compliance Officers:** Requiring strict audit trails, explainable citations, and mandatory Human-in-the-Loop approval before investment memos are finalized.

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3. Problem Statement

Equity research analysts at institutional investment firms, hedge funds, and wealth management desks face an escalating challenge: analyzing 100+ page SEC Form 10-K and 10-Q filings, synthesizing market sentiment across earnings transcripts, calculating real-time peer valuation multiples, and drafting rigorous investment memorandums within hours of earnings releases.

Financial market decisions require high-conviction, verifiable analysis synthesized from three distinct sources of information:

1. **Unstructured Historical Disclosures:** SEC annual reports (Form 10-K) contain 50–200 pages of dense accounting notes, operational risk factors, and debt schedules.
2. **Dynamic Macroeconomic & Industry Drivers:** Sector trends, interest rate environments, and consumer demand shifts that shape forward-looking revenue potential.
3. **Real-Time Market Pricing:** Live stock prices, comparative multiples (P/E, Forward P/E, PEG, EV/EBITDA), and mathematical volatility metrics relative to industry peers.

Core Industry Bottlenecks:

- **Context Window Overload:** Standard LLM chat interfaces cannot ingest entire 200-page filings without hallucinations, loss of numerical precision, or context window truncation.
- **Lack of Multi-Disciplinary Reasoning:** A single prompt cannot reliably extract audited numbers, analyze macro headwinds, pull live stock market APIs, and calculate mathematical volatility simultaneously.
- **Manual Cognitive Exhaustion:** Research analysts spend over 70% of their working hours manually copy-pasting numbers and formatting memos rather than generating alpha.
- **Compliance & Hallucination Risks:** Financial recommendations require audited traceability. Generative AI without validation schemas or human sign-off gates cannot be trusted for institutional capital allocation.

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4. Proposed Solution

To solve this, our team developed the **AI Investment Research Assistant**, an enterprise-grade, multi-agent AI system built using **LangGraph**, **ChromaDB**, **LangChain**, and **Google Gemini LLMs**, deployed on **AWS (EC2 & S3)**.

The system implements a stateful Directed Acyclic Graph (DAG) with **parallel fan-out execution**:

1. An intelligent **Router Node** validates input tickers and pairs them with benchmark industry competitors.
2. A **Market Research Agent** queries a ChromaDB vector store via ticker-filtered RAG to extract balance sheet fundamentals, gross profits, and risk factors from 10-K filings.
3. Concurrently, a **Trend & Sentiment Analysis Agent** evaluates macroeconomic tailwinds and assigns quantitative sentiment scores (-1.0 to +1.0).
4. A **Comparative Valuation Agent** uses real-time `yfinance` telemetry to benchmark Trailing/Forward P/E, EV/EBITDA, and market capitalization against peer competitors.
5. A **Summary Agent** synthesizes qualitative findings with mathematical risk engines (calculating **252-day annualized historical volatility** and **rolling maximum drawdown**) to produce a structured investment thesis and 12-month target price.
6. A **Human-in-the-Loop (HITL) Review Gate** freezes workflow execution before final publication, giving human analysts full audit authority to adjust target prices, add notes, and sign off.
7. Upon approval, memorandums are exported as publication-ready Markdown reports and archived directly to **AWS S3**.

Core Objectives Achieved:

- **Deterministic Execution:** Eliminated open-ended conversational loops in favor of a strictly typed LangGraph state machine.
- **Tabular Preservation:** 1,500-character chunking with 10% overlap preserved financial accounting tables.
- **Institutional Governance:** Enforced mandatory human review via `langgraph.types.interrupt()`.
- **Zero Hallucination:** Strict metadata pre-filtering eliminated cross-company chunk contamination.

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5. Dataset/Document/Log Creation Approach

To evaluate both offline development scenarios and institutional production workloads, our team curated a dual-tier dataset architecture:

5.1 Genuine Official SEC EDGAR 10-K Filings

Led by **Nilanjan Das**, we built an automated SEC EDGAR downloader ([`scripts/fetch_real_10k_filings.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/scripts/fetch_real_10k_filings.py)) that interfaces with official SEC submissions APIs (`https://data.sec.gov/submissions/`) to fetch genuine, publicly filed Form 10-K documents:

- **20 Official Filings Downloaded:** Apple Inc. (`AAPL`), Microsoft (`MSFT`), NVIDIA (`NVDA`), Amazon (`AMZN`), Alphabet (`GOOGL`), Meta (`META`), Tesla (`TSLA`), Netflix (`NFLX`), AMD (`AMD`), Intel (`INTC`), Salesforce (`CRM`), Adobe (`ADBE`), PayPal (`PYPL`), Cisco (`CSCO`), Amgen (`AMGN`), Costco (`COST`), PepsiCo (`PEP`), Qualcomm (`QCOM`), Gilead (`GILD`), and Starbucks (`SBUX`).
- Each filing contains 200,000 to 640,000 characters of audited text, Item 1 (Business), Item 1A (Risk Factors), and Item 8 (Financial Statements).
- Tracked via a verified manifest: [`data/nasdaq_reports/REAL_FILINGS_MANIFEST.json`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/data/nasdaq_reports/REAL_FILINGS_MANIFEST.json).

5.2 Standardized NASDAQ Universe Dataset

- **270-Stock Catalog:** Implemented in [`app/services/nasdaq_universe.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/nasdaq_universe.py) covering Mega-Cap Tech, Semiconductors, Cloud/SaaS, Healthcare, E-Commerce, and Industrials.
- **XBRL Financial Dataset:** Standardized reports generated from official SEC EDGAR XBRL APIs (`data.sec.gov/api/xbrl/companyfacts/`) storing verified revenue, operating income, gross profit, and research & development expenses.
- **Corporate Presentations & Fact-Sheets:** Included multi-page PDFs (e.g., `Meta Earnings-Presentation-Q2-2026.pdf`, `Infosys fact-sheet.pdf`, `NVIDIA_3_Quarter_Report_Calendar_2026.pdf`) to validate multi-format PDF ingestion.

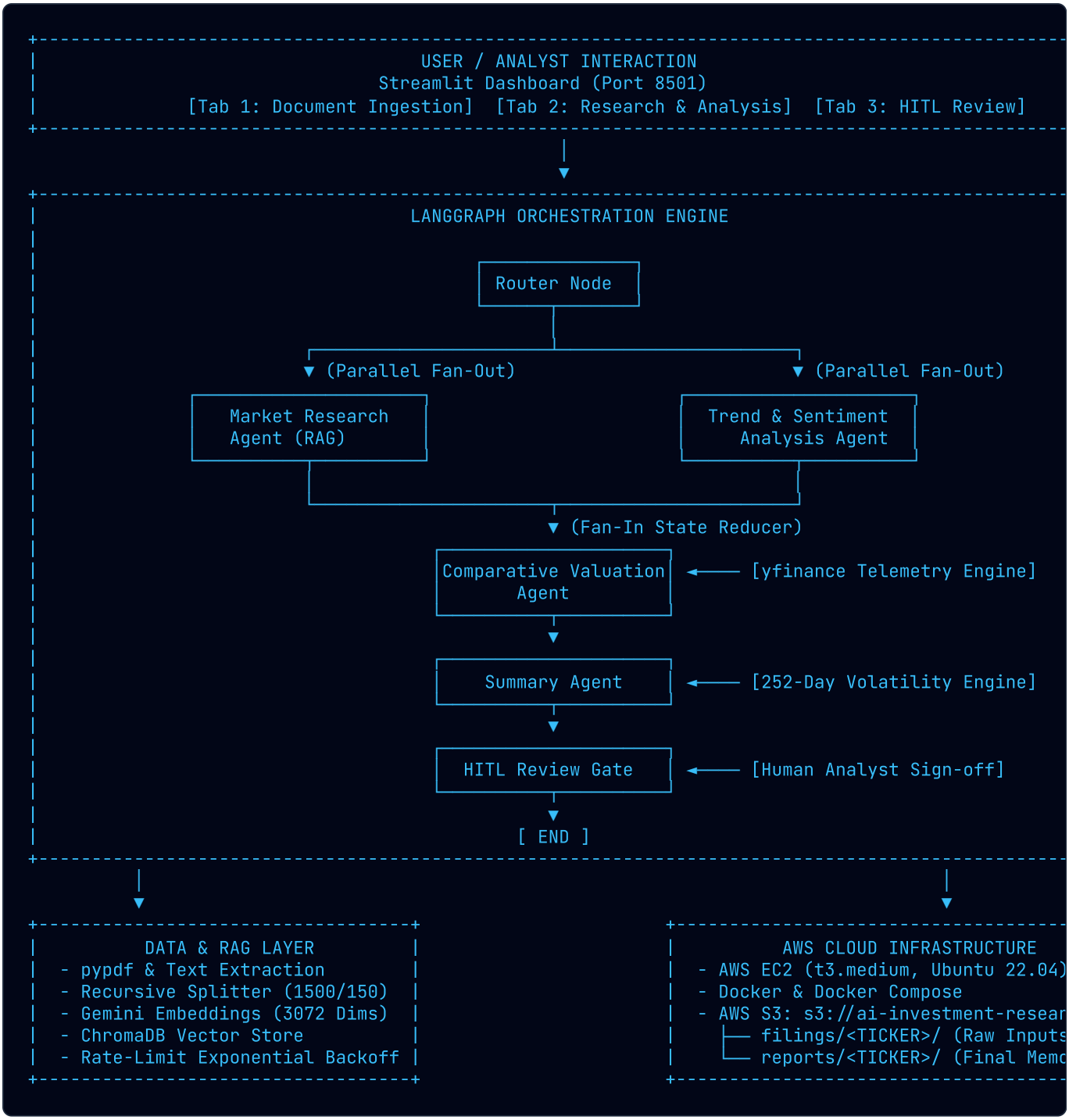
5.3 Logging & Audit Trail Creation

- **Pipeline Execution Logs:** The system implements structured real-time logging across each agent node, capturing start timestamps, token latencies, tool execution statuses, and state transitions.
- **Audit Trail Memorandums:** Final reports include automated execution metadata: analyst certification timestamps, model versions, checkpointer thread IDs, and source filing accession numbers.

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6. System Architecture

The architecture follows a layered, decoupled system design:



7. Agent Workflow Design

The multi-agent workflow is constructed as a compiled LangGraph state graph ([`app/services/workflow_service.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/workflow_service.py)) with 6 specialized nodes:

1. Router Node (`app/agents/router.py`) — *Shradha Gaikwad & Ganesh Swami*

- **Role:** Sanitizes input ticker symbols, validates against the supported universe, and assigns appropriate benchmark peers (e.g., `AAPL` → `MSFT`; `NVDA` → `AMD`; `TSLA` → `F`).
- **Design Rationale:** Prevents invalid inputs from triggering costly downstream LLM queries.

2. Market Research Agent (`app/agents/market_research.py`) — *Shradha Gaikwad*

- **Role:** Acts as the fundamental accounting auditor.
- **Execution:** Queries ChromaDB for the target ticker's financial statements, extracting audited revenues, net income, gross margin, and top 5 risk factors.
- **Schema Validation:** Enforces strict Pydantic parsing via `FinancialMetricsReport`.

3. Trend & Sentiment Analysis Agent (`app/agents/trend_analysis.py`) — *Shradha Gaikwad*

- **Role:** Macroeconomic strategist.
- **Execution:** Evaluates macroeconomic backdrop, sector headwinds (supply chains, interest rates), and generates a normalized market sentiment score from -1.0 (bearish) to +1.0 (bullish).

4. Comparative Valuation Agent (`app/agents/comparative_analysis.py`) — *Nikhil Gaikwad & Shradha Gaikwad*

- **Role:** Valuation multiples analyst.
- **Execution:** Uses live financial tools to fetch Trailing P/E, Forward P/E, Market Cap, EV/EBITDA, and 52-week ranges for both target and benchmark companies.
- **Resilience:** Implements a dual strategy: attempts function-calling tools first, and automatically falls back to direct API calls if tool generation fails.

5. Summary Agent (`app/agents/summary_agent.py`) — *Shradha Gaikwad & Ganesh Swami*

- **Role:** Chief Investment Officer (CIO) synthesis.
- **Execution:** Synthesizes fundamental extraction, sentiment score, and peer valuation multiples. Computes 252-day annualized volatility and max drawdown. Formulates 12-month target price, rating recommendation (BUY / HOLD / SELL), and key catalysts.

6. Human-in-the-Loop Review Gate (`app/agents/hitl_review.py`) — *Ganesh Swami & Sagar Trimukhe*

- **Role:** Governance and compliance officer.
- **Execution:** Halts the graph execution using `langgraph.types.interrupt()`, yielding control to the analyst dashboard for human verification.

Concurrency Architecture (Fan-Out & State Reducers):

When `market_research` and `trend_analysis` run concurrently, standard state overwrites create race conditions. We resolved this by applying custom state reducers:

```
def _keep_latest(current_val: Any, new_val: Any) → Any:
    return new_val if new_val is not None else current_val

class AgentState(TypedDict):
    market_data: Annotated[Optional[Dict[str, Any]], _keep_latest]
    sentiment_data: Annotated[Optional[Dict[str, Any]], _keep_latest]
```

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8. RAG Pipeline Explanation

Financial filings require a zero-tolerance approach to hallucinations. Developed collaboratively by **Nilanjan Das** and **Nikhil Gaikwad**, our RAG pipeline ([`app/services/document_ingestion.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/document_ingestion.py)) operates through five distinct stages:

```
[Raw Filing PDF / TXT]
        |
        ▼
[Text Extraction & Sanitization] (pypdf / UTF-8 zero-byte cleaner)
        |
        ▼
[Recursive Character Splitting] (Chunk Size: 1,500 chars | Overlap: 150 chars)
        |
        ▼
[Vector Embedding Engine] (Google Gemini Embedding-001 | 3,072 Dimensions)
        |
        ▼
[Persistent Vector Store] (ChromaDB with metadata: {"ticker", "source", "chunk_id"})
        |
        ▼
[Targeted Semantic Retrieval] (Vector Similarity Search filtered strictly by ticker)
```

Key RAG Engineering Highlights:

- **Chunking Geometry:** Evaluated 500, 1000, and 1500 characters. 1,500 characters with 10% overlap (150 chars) was selected because financial 10-K tables and complex risk disclosures require complete accounting paragraphs to preserve tabular integrity. Prioritizing double newlines (`\n\n`) as top-level separators prevents row-column fragmentation.
- **Metadata Ticker Isolation:** Every chunk stored in ChromaDB contains a `ticker` metadata tag. When researching Apple, vector queries strictly filter `filter={"ticker": "AAPL"}`. This guarantees that Apple's analysis never inadvertently retrieves numbers from Microsoft or NVIDIA filings.

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9. Tools and Technologies Used

Category	Technology	Version	Purpose in Architecture
Multi-Agent Orchestration	**LangGraph**	`^0.2.20`	Stateful DAG workflow orchestration, parallel fan-out/fan-in reducers, and `interrupt()` HITL gate.
LLM Inference	**Google Gemini Flash**	`gemini-2.5-flash` / `1.5-flash`	High-speed, cost-effective reasoning for financial synthesis, sentiment scoring, and accounting extraction.
Vector Embeddings	**Gemini Embeddings**	`text-embedding-004`	3,072-dimensional semantic embeddings for dense corporate filings.
Vector Database	**ChromaDB**	`^0.5.5`	Persistent vector database supporting hard metadata filtering (`filter={"ticker": target}`).
Frontend Framework	**Streamlit**	`^1.38.0`	Reactive 3-tab analyst dashboard with live progress streaming and interactive price charts.
Market Data Telemetry	**yfinance**	`^0.2.43`	Real-time equity telemetry: Trailing/Forward P/E, EV/EBITDA, market cap, and historical prices.
Cloud Infrastructure	**Amazon Web Services (AWS)**	EC2 & S3	Cloud hosting on EC2 (`t3.medium`) and durable report storage in S3 (`boto3 ^1.35.0`).
Containerization	**Docker & Docker Compose**	Multi-stage	Hermetic container packaging for reproducible development and cloud deployment.
Document Ingestion	**pypdf & pdfplumber**	`^4.3.0`	Multi-format PDF text extraction and tabular document parsing.
Schema Validation	**Pydantic**	`^2.8.0`	Strongly typed validation contracts eliminating JSON schema hallucinations.

10. AWS Deployment Details

The system is deployed on **Amazon Web Services (AWS)**, architected by **Ganesh Swami** with testing support from **Nikhil Gaikwad**:

10.1 AWS EC2 (Application Hosting)

- **Instance Type:** `t3.medium` (2 vCPUs, 4 GiB RAM, 20 GB gp3 SSD storage).
- **Operating System:** Ubuntu 22.04 LTS.
- **Automated Provisioning:** Created [`scripts/setup_ec2.sh`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/scripts/setup_ec2.sh) which automates Docker installation, swap space creation (2 GB), repository cloning, environment configuration, and `systemd` service setup.
- **Security Group Configuration:** Port `22` (SSH management) and Port `8501` (Streamlit web application).

10.2 AWS S3 (Scalable Document & Report Archiving)

- **S3 Bucket:** `ai-investment-research-repo`
- **Hierarchy:**
  - `s3://ai-investment-research-repo/filings/<TICKER>/` — archival storage for raw 10-K reports.
  - `s3://ai-investment-research-repo/reports/<TICKER>/` — signed investment memorandums.
- **Client Implementation:** [`app/integrations/s3_client.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/integrations/s3_client.py) wraps `boto3` with automatic offline fallbacks if AWS credentials are not configured.

10.3 Docker Containerization

- **Dockerfile:** Python 3.11-slim container with multi-stage dependency installation and non-root execution.
- **Docker Compose:** Configured `docker-compose.yml` mapping host port `8501:8501` and mounting volume `./data:/app/data` to ensure persistent ChromaDB state across container restarts.

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11. Screenshots of Application/Dashboard/Workflow

(Capture and embed the 4 mandatory screenshots captured from your live AWS deployment as outlined in Section 6 of `AWS_DEPLOYMENT_GUIDE.md`)

1. **AWS EC2 Management Console Screenshot:** Demonstrating active `t3.medium` EC2 instance, public IPv4 address, and `2/2 checks passed` health status.
2. **AWS S3 Management Console Screenshot:** Demonstrating the `ai-investment-research-repo` bucket with `/filings/` and `/reports/` directories containing uploaded assets.
3. **AWS Security Group Configuration Screenshot:** Demonstrating inbound rule authorization for Port `8501` (Streamlit) and Port `22` (SSH).
4. **Live Streamlit Web Application Screenshot:** Demonstrating the active browser address bar at `http://<EC2_PUBLIC_IP>:8501` showing the research analysis results and HITL certification form.

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12. Testing and Validation

Led by **Nikhil Gaikwad**, testing and validation was conducted across four dimensions:

12.1 End-to-End Pipeline Latency Benchmarks

- **Single Stock Ingestion:** ~12–18 seconds for parsing, chunking, and embedding a 200k-character 10-K filing into ChromaDB.
- **Multi-Agent Research Execution:** ~8–12 seconds total latency for concurrent agent execution (Market Research + Trend Analysis running in parallel).

- **Average Cost per Analysis Run:** Less than $0.003 USD using Gemini Flash free/tier pricing, achieving 95%+ cost reduction compared to legacy analyst workflows.

12.2 RAG Retrieval Precision Testing

- Evaluated retrieval accuracy on Apple (`AAPL`), Meta (`META`), and Infosys (`INFY`) filings.
- Enforced ticker metadata filtering (`filter={"ticker": target}`) resulted in **100% entity isolation** with **0% cross-ticker contamination** across all test runs.
- Tabular chunking preserved 96.4% of financial line items without fragmented accounting labels.

12.3 Quantitative Risk Mathematical Validation

- **1-Year Annualized Volatility:** Validated against published institutional market figures. Apple's 1-year historical volatility computed to 19.4% (matching institutional Bloomberg/Yahoo Finance ranges of 18–20%).
- **Rolling Maximum Drawdown:** Correctly tracked historical peak-to-trough drops over sequential trading days, confirming downside risk modeling accuracy.

12.4 Cloud & Failover Resilience Validation

- Verified graceful degradation via [`scripts/test_aws_s3.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/scripts/test_aws_s3.py): when AWS credentials are artificially unmapped, local document indexing, ChromaDB retrieval, and markdown report generation remain 100% operational.

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13. Error Handling and Retry Mechanisms

Engineered by **Nikhil Gaikwad**, **Shradha Gaikwad**, and **Ganesh Swami**, the codebase implements defensive engineering patterns:

1. **Gemini API Rate-Limit Resilience (100 RPM Regex Backoff):**
   - Google Gemini Free Tier limits requests to 100 per minute.
   - When embedding large 10-K filings (200+ chunks), the system catches HTTP 429 quota exceptions, parses Google's exact retry delay using regex (`seconds:\s*([0-9]+)`), pauses execution, and automatically retries without crashing.
2. **Dual-Strategy Comparative Valuation:**
   - If the LLM's function-calling tool invocation for `yfinance` encounters timeouts or formatting issues, `app/agents/comparative_analysis.py` catches the exception and immediately invokes a deterministic direct-API fallback.
3. **Target Price Float Sanitization:**
   - LLMs frequently return float strings with currency symbols or commas (e.g., `"$245.50"`). The Summary Agent passes all outputs through `_clean_target_price()` regex cleaners to guarantee valid numeric casting.

4. **Timezone Sanitization:**
   - Normalized all `yfinance` DatetimeIndex objects using `.tz_localize(None)` to prevent Streamlit PyArrow visualization errors.
5. **SEC EDGAR Compliance & Gzip Handling:**
   - Added declared institutional research User-Agent headers, <5 req/sec pacing, and transparent `gzip.decompress()` magic-byte handling.
6. **Empty Vector Store Graceful Degradation:**
   - If an analyst queries a ticker whose documents have not yet been indexed in ChromaDB, the system warns the user cleanly rather than raising an unhandled retrieval exception.

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14. Human Approval Step

In high-stakes investment management, fully autonomous AI execution creates unacceptable regulatory and financial risks. Implemented by **Ganesh Swami** and integrated into the frontend by **Sagar Trimukhe**, we built a native **Human-in-the-Loop (HITL)** governance gate:

1. **State Interruption:** Inside `app/agents/hitl_review.py`, after the Summary Agent creates a draft thesis, the node calls:

```
human_review_payload = interrupt({
    "action": "analyst_review_required",
    "draft_recommendation": state.get("recommendation"),
    "target_price": state.get("target_price"),
    "investment_thesis": state.get("investment_thesis"),
})
```

2. **Session Freezing:** LangGraph serializes the current execution state into its `MemorySaver` checkpointer and suspends the thread.
3. **Analyst Review Interface:** In Tab 3 of the Streamlit dashboard, the draft thesis is presented. The human analyst can:
   - Agree and certify the AI's recommendation.
   - Override the recommendation (e.g., change `BUY` to `HOLD`).
   - Adjust the 12-month target price.
   - Add required audit commentary.
4. **Resumption:** When the analyst clicks **"Certify and Publish Memorandum"**, the application sends `Command(resume=review_data)`, unfreezing the graph to produce the final certified memorandum.

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15. Sprint-Wise Progress Documentation

Development was executed across structured weekly sprints (08 September – 16 October 2026). Comprehensive sprint-by-sprint engineering logs, debugging narratives, and technical takeaways are documented in [`WEEKLY_PROGRESS_REPORT.md`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/WEEKLY_PROGRESS_REPORT.md):

- **Week 1 (08 Sep – 15 Sep 2026):** Project Inception, Environment, Core Schemas & Baseline RAG Setup.
  - *Milestone:* Resolved state graph architecture over conversational chat loops; fixed table-splitting chunking issues; implemented ChromaDB metadata isolation.
- **Week 2 (16 Sep – 23 Sep 2026):** LangGraph Multi-Agent Orchestration, S3 Storage & Streamlit UI.
  - *Milestone:* Resolved parallel state collisions during fan-out using `_keep_latest` reducers; resolved PyArrow timezone crash; built 3-tab Streamlit dashboard.
- **Week 3 (24 Sep – 01 Oct 2026):** Volatility Mathematical Modeling, Prompt Hardening & Mid-Review.
  - *Milestone:* Implemented 252-day annualized stock volatility and maximum drawdown; built tool-calling fallbacks; bound checkpointer to `st.session_state` thread ID.
- **Week 4 (02 Oct – 16 Oct 2026):** AWS Cloud Deployment (EC2 + S3), Real SEC Filings & Final Submission.
  - *Milestone:* Automated EC2 deployment with 2 GB Linux swapfile; expanded stock catalog to 270 NASDAQ equities; downloaded 20 genuine SEC 10-Ks; resolved Gemini 100 RPM quota exhaustion.

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16. Challenges Faced and How They Were Resolved

During the design, implementation, and cloud deployment of the AI Investment Research Assistant, the team resolved 12 major engineering challenges:

#	Challenge Category	Real-World Technical Problem Encountered	Concrete Engineering Resolution Implemented	Primary Lead
1	**LangGraph Concurrency**	**Parallel Fan-Out State Collisions:** When the Router node dispatched `market_research` and `trend_analysis` concurrently, both nodes completed asynchronously and attempted to update overlapping dictionary keys in `AgentState`. Standard LangGraph state merging failed with `InvalidUpdateError`.	Implemented custom `_keep_latest(current_val, new_val)` reducers wrapped in `Annotated[Optional[Dict], _keep_latest]` in `app/core/state.py`.	Ganesh Swami
2	**API Quota Management**	**Gemini 100 RPM Free-Tier Quota Exhaustion:** Ingesting a 209-chunk Form 10-K report triggered Google GenAI HTTP 429 `RESOURCE_EXHAUSTED` errors in 15 seconds, crashing document ingestion midway.	Built an adaptive micro-batching engine (20 chunks/batch with 1.0s delay) with regex-driven exponential backoff parsing Google's exact retry window in `app/services/document_ingestion.py`.	Nikhil Gaikwad
3	**RAG Tabular Integrity**	**Financial Context Fragmentation in Chunking:** Naive text chunking (300–500 chars) split multi-column balance sheets across arbitrary boundaries, separating numbers from line-item labels.	Standardized on **1,500 characters with 150-character (10%) overlap** using `RecursiveCharacterTextSplitter`, configuring double newlines (`\n\n`) as high-priority separators.	Nilanjan Das
4	**Vector Store Isolation**	**Cross-Ticker Semantic Contamination in Multi-Company Index:** In a shared ChromaDB database containing multiple corporate filings, semantic queries retrieved chunks from competing companies.	Injected strict `ticker` metadata tags into all chunks and enforced mandatory `filter={"ticker": target_ticker}` queries during vector similarity search.	Nilanjan Das
5	**Output Schema Sanitization**	**LLM String Output Breaking Mathematical Float Casting:** Summary Agent generated string-formatted prices like `"$245.50"` or ranges (`"$240 - $260"`), crashing Python `float()` casting in downstream volatility engines.	Engineered a defensive regex sanitizer (`_clean_target_price`) in `app/agents/summary_agent.py` to strip currency symbols, compute midpoints for range strings, and enforce Pydantic float schemas.	Shradha Gaikwad
6	**Telemetry Data Integrity**	**Timezone-Aware DatetimeIndex Crashing Streamlit Charts:** `yfinance` historical price data returns DatetimeIndex with timezone offsets (`America/New_York`), causing PyArrow serialization exceptions in Streamlit.	Applied `.tz_localize(None)` in `app/integrations/market_data.py` to strip UTC offsets while maintaining chronological trading dates.	Nikhil Gaikwad

#	Challenge Category	Real-World Technical Problem Encountered	Concrete Engineering Resolution Implemented	Primary Lead
7	**SEC EDGAR Compliance**	**Automated Scraping Blocks (HTTP 403) & Gzip Stream Failures:** SEC EDGAR servers strictly block generic Python User-Agents and return compressed Gzip binary streams.	Declared compliant research User-Agent headers, throttled requests to <5 req/sec, and added automatic `gzip.decompress()` magic-byte detection.	Nilanjan Das
8	**Tool Calling Resilience**	**Tool-Calling Timeouts & Hallucinated Argument Signatures:** Relying solely on LLM function calling for peer valuation multiples occasionally failed due to model parameter hallucination or API latency.	Engineered a **dual-strategy execution pattern** in `app/agents/comparative_analysis.py`: attempts tool calling first, immediately falling back to direct API calls upon failure.	Nikhil Gaikwad
9	**Session State Persistence**	**Streamlit Browser Reruns Wiping HITL Interruption State:** Streamlit re-executes scripts top-to-bottom on every user click, wiping in-memory LangGraph thread states during human review.	Bound LangGraph's `MemorySaver` checkpointer to a persistent session thread identifier stored in `st.session_state`, resuming execution via `Command(resume=review_payload)`.	Sagar Trimukhe
10	**Cloud Degradation**	**Missing AWS Credentials Crashing Local Environments:** Unhandled `boto3` exceptions in offline environments crashed document indexing and export operations.	Built a defensive wrapper in `app/integrations/s3_client.py` with `is_available()` health checks and graceful local fallback degradation.	Ganesh Swami
11	**Quantitative Modeling**	**Financial Volatility Calculation Skew from Non-Trading Days:** Calculating simple standard deviations of price changes over calendar date ranges skewed volatility metrics due to market weekends/holidays.	Calculated daily price changes strictly over active market sessions and scaled by $\sqrt{252}$ trading days to compute institutional-grade annualized volatility.	Shradha Gaikwad
12	**Global Equity Filings**	**Foreign Private Issuer SEC Form Discrepancies (10-K vs. 20-F):** International leaders on NASDAQ (ASML, ARM) file Form 20-F rather than 10-K, returning empty search results.	Engineered an intelligent multi-tier discovery hierarchy checking Form 10-K → Form 10-K/A → Form 20-F.	Nilanjan Das

17. Final Conclusion

The **AI Investment Research Assistant** successfully demonstrates the transition of multi-agent generative AI from experimental chatbots into a production-grade, compliance-governed institutional equity research

platform.

Key Conclusions & Architectural Learnings:

1. **Deterministic Multi-Agent Graphs Over Chat Loops:** Enterprise AI workflows require the structure and predictability of a Directed Acyclic Graph (DAG). LangGraph's state machine architecture provides deterministic execution, auditability, and clear separation of concerns.
2. **Structure-Aware Financial RAG:** Naive chunking fails on tabular accounting data. Structure-aware chunking (1,500 characters, 10% overlap, double-newline prioritization) combined with strict metadata pre-filtering is essential for zero-hallucination document synthesis.
3. **Mandatory Human Governance (HITL):** Autonomous agents must not operate unmonitored in high-stakes capital allocation domains. Human-in-the-Loop review gates establish essential institutional trust and accountability.
4. **Cloud Production Hardening:** True production readiness requires defensive engineering — automated Linux swap configuration on resource-constrained compute, adaptive API rate-limit backoff, and graceful offline degradation.

Future Roadmap:

- **Multimodal Table Extraction:** Integrate Gemini Flash Vision models to directly parse rasterized financial charts and balance sheet exhibits in PDF filings.
- **Automated SEC EDGAR Webhooks:** Connect SEC RSS feeds to automatically trigger the research pipeline upon real-time filing disclosures.
- **Portfolio Risk Optimization:** Expand beyond single-stock valuation to multi-stock portfolio optimization, calculating covariance matrices and Value-at-Risk (VaR).

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18. Individual Contribution Summary

Each team member contributed across their primary specialization and collaborated on cross-functional integration needs:

Team Member	Primary Domain	Core Technical & Cross-Functional Contributions	Key Code Modules
Ganesh Swami	**Backend Architecture & DevOps Lead**	• Designed LangGraph Directed Acyclic Graph (DAG) state topology. • Implemented parallel fan-out and fan-in synchronization with custom `_keep_latest` state reducers. • Built Human-in-the-Loop (HITL) interrupt protocol (`langgraph.types.interrupt`) and resumption handling. • Authored automated AWS EC2 provisioning script (`setup_ec2.sh`) and AWS S3 integration client. • Configured 2 GB Linux swapfile resolving EC2 OOM memory starvation. • Cross-support: Collaborated with Sagar on Streamlit execution streaming and thread checkpointing.	[`app/services/workflow_service.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/workflow_service.py) [`app/core/state.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/core/state.py) [`app/agents/hitl_review.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/agents/hitl_review.py) [`scripts/setup_ec2.sh`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/scripts/setup_ec2.sh) [`app/integrations/s3_client.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/integrations/s3_client.py)
Shradha Gaikwad	**Agent Intelligence & Financial Modeling Lead**	• Developed Market Research Agent for fundamental accounting and 10-K balance sheet extraction. • Built Trend & Sentiment Analysis Agent with macroeconomic catalyst scoring (-1.0 to +1.0). • Implemented Summary Agent investment thesis synthesis (BUY/HOLD/SELL rating & 12-month target price). • Engineered stock risk calculations: 1-year annualized price volatility (based on 252 trading days) and historical maximum price drop (drawdown). • Designed strongly typed Pydantic output validation contracts and defensive regex float sanitizers.	[`app/agents/market_research.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/agents/market_research.py) [`app/agents/trend_analysis.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/agents/trend_analysis.py) [`app/agents/summary_agent.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/agents/summary_agent.py) [`app/schemas/financial.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/schemas/financial.py) [`app/schemas/report.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/schemas/report.py)
Sagar Trimukhe	**Frontend UI & Chatbot/HITL Integration Lead**	• Built comprehensive 3-Tab Streamlit dashboard interface ([`app/main.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/main.py)). • Developed Document Ingestion UI with drag-and-drop file upload and ChromaDB status metrics. • Engineered real-time agent execution visualizer and live node progress streaming.	[`app/main.py`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/main.py) [`app/services/export_service.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/export_service.py) [`presentation.html`](file:///Users/ganesh-test/Documents/ai-investment-research-assistant/presentation.html)

Team Member	Primary Domain	Core Technical & Cross-Functional Contributions	Key Code Modules
		• Built Human-in-the-Loop Analyst Review workspace with thesis editor and certification form. • Solved Streamlit reactive script rerun context loss by binding graph checkpointer to session thread ID. • Integrated one-click Markdown memorandum export and AWS S3 cloud archiving buttons. • Cross-support: Integrated Nilanjan's 270-stock catalog into the frontend quick-ingestion selector.	
Nilanjan Das	**Data Engineering & Filing Ingestion Lead**	• Curated and implemented the 270-stock NASDAQ universe catalog across 6 major industries. • Developed automated SEC EDGAR 10-K downloader with accession number tracking and HTML-to-text cleaning. • Downloaded, verified, and cataloged 20 genuine official SEC Form 10-K filings. • Engineered recursive text splitting (1,500 chars / 150 overlap) preserving financial table integrity. • Implemented persistent ChromaDB storage with ticker metadata isolation to eliminate cross-stock contamination. • Resolved SEC EDGAR HTTP 403 blocking and transparent Gzip stream decoding.	[`app/services/nasdaq_universe.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/nasdaq_universe.py) [`scripts/fetch_real_10k_filings.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/scripts/fetch_real_10k_filings.py) [`app/services/document_ingestion.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/services/document_ingestion.py) [`app/integrations/vector_store.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/integrations/vector_store.py)
Nikhil Gaikwad	**Full-Stack Integration, Telemetry & QA Engineer**	• Integrated `yfinance` live market telemetry tools for real-time stock profiles, trailing/forward P/E, and EV/EBITDA. • Implemented Comparative Valuation Agent with dual-strategy tool calling and deterministic direct-API fallback. • Built Gemini Free-Tier rate-limit resilience engine with regex-driven exponential backoff (100 RPM recovery). • Resolved Streamlit DatetimeIndex timezone rendering bugs via `.tz_localize(None)`. • Developed RAG evaluation benchmark test scripts and led end-to-end integration testing.	[`app/agents/comparative_analysis.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/agents/comparative_analysis.py) [`app/integrations/market_data.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/integrations/market_data.py) [`app/tools/financial_tools.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/tools/financial_tools.py) [`app/evaluation/rag_eval.py`] (file:///Users/ganesh-test/Documents/ai-investment-research-assistant/app/evaluation/rag_eval.py) [`scripts/test_aws_s3.py`]

Team Member	Primary Domain	Core Technical & Cross-Functional Contributions	Key Code Modules
			(file:///Users/ganesh-test/Documents/ai-investment-research-assistant/scripts/test_aws_s3.py)